

Design and operational issues in AGV-served manufacturing systems

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Automated Guided Vehicle (AGV) systems are already in widespread use and their importance for material handling is expected to grow rapidly. The advantages that such systems can offer include increased flexibility, better space utilization, improved factory floor safety, reduction in overall operating cost, and easier interface with other automated systems. This survey paper focuses on design and operational issues that arise in AGV systems. The objectives of the paper are to unify various lines of research related to AGVs and to suggest directions for future study. We consider problems arising in flowpath design, fleet sizing, job and vehicle scheduling, dispatching and conflict-free routing. Flowpath design problems address computationally intractable issues in the physical layout of a single loop and complex networks. Transportation and related models, waiting line analysis and simulation approaches are used to address fleet sizing questions. Scheduling issues focus on three flowpath layouts. In line layouts, the most important issues include finding an efficient job sequencing algorithm and identifying optimal AGV launch times. In loop layouts, issues such as joint scheduling of the job and AGV schedules, interface with a larger manufacturing system, dynamic job arrivals, and the location of the AGV parking area, are important. For complex network layouts, joint scheduling, heuristic dispatching rules, and conflict-free routing of AGVs, are considered. We identify the inefficiencies that result from addressing these issues in isolation, suggesting the need for integration. We also provide a summary of the most important open research issues related to all the above topics.

Keywords: manufacturing, automated guided vehicles, design and operation, survey

1. Introduction

Automated Guided Vehicle (AGV) systems are an important part of many low to medium volume manufacturing operations, including flexible manufacturing systems (FMSs), warehousing, and service industries where they are used for moving jobs as diverse as mail, laundry and hospital meals. AGVs are battery-powered driverless

vehicles, centrally computer-controlled and independently addressable. They move either along wire guidepaths, or by magnetic or optic guidance. They are used to move jobs between workstations on a factory floor. The relatively inexpensive guidepath which does not interfere with other material flows, and several other advantages discussed below over alternative systems, make AGV systems an important material handling method of the future.

AGVs can offer many advantages over push carts, forklifts or fixed material handling devices such as conveyors. The primary advantages occur in flexibility, space utilization, safety and overall operating cost. AGV systems are extremely flexible, since the flowpath (also called guidepath) can be modified in a matter of minutes, and even wire-guided vehicles can be dynamically rerouted to respond to changing priorities within an existing system. Unlike conveyors, AGVs only occupy workspace when working temporarily in a given area, or when parked. Since they do not create physical barriers within the factory as conveyors do, AGV systems can share space with other uses such as pedestrian or forklift aisles, thus improving overall space utilization within the factory. AGVs are safer than competitive technologies: all vehicles are equipped with lights and horns to warn pedestrians of their presence. Unlike forklifts, the possibility of operator error is removed. Although they are significantly more expensive than forklifts or other material handling equipment, AGV systems pay for themselves rapidly by reducing unit operating costs, especially in multiple shift operations.

The first large-scale manufacturing application of an AGV system occurred in 1974 at a Volvo plant in Kalmar, Sweden. Within little more than a decade thereafter, about 3,300 plants worldwide employed more than 15,000 AGVs [186]. The largest application in North America is at the truck assembly plant of General Motors (GM) in Oshawa, Canada [56], where 1,012 AGVs transport truck engines, bodies, and chassis across the 2.7 million square feet plant. Other large-scale applications include 135 AGVs that carry autobodies through dozens of automated welding stations, sealing stations, and manual work stations at GM's Doraville, Georgia plant [58], and 60 AGVs being used to transport films, paper and chemicals at Eastman Kodak's main distribution center in Rochester, New York [115]. *The New York Times* employs 23 AGVs to transport newsprint from the raw materials warehouse to the roll lay down and preparation area, which facilitates printing 45,000 newspapers an hour by 6 color presses [6]. Although these examples illustrate the diverse applications of AGVs in North America, at this time Japan and Europe lead in AGV system applications. In 1989, for example, Japanese companies purchased 5,000 vehicles, European companies 3,000 vehicles, and US companies only 500 vehicles.

Articles that describe industrial applications and those that review the state-of-the-art in AGV systems appear regularly in trade publications such as *Modern Materials Handling* and *Material Handling Engineering*. The interested reader may refer to Gould [57,59], Schwind [148] and Forger [40]. The usefulness of AGV systems implementations is discussed in the July 1990 issue of *Modern Materials*

Handling [130]. It is reported that 56% of all AGV system installations in 1989 were in just-in-time (JIT) systems. D. Butler of Caterpillar Corporation is quoted as follows: "We expect these JIT applications to continue as a major use of AGV systems. The opportunities for cost savings when reducing inventories do not stop with reduced carrying costs and lower capital investment. Savings in reduced floor space, insurance costs, indirect labor, storage equipment, and many other areas also can be realized."

The mechanical design of AGVs and their guidance methods are usually studied by mechanical and electrical engineers. See, for example, Koff [97] and Vosniakos and Mamalis [179]. An excellent buyer's guide to AGVs appears in the *Industrial Engineering Solutions* issue of April 1996 [75]. This guide lists the names of AGV manufacturers, vehicle types, guidance methods used, communication systems, load capacities of AGVs, travel speeds, battery types, price ranges and other details. The *design and operation of AGV systems*, however, must be of direct concern to the OR/MS community.

The design of AGV systems involves flowpath design and fleet size determination. The different flowpath configurations [101] for AGVs include single line (figure 1), single loop (figure 2), ladder type (figure 3), and complex network (figure 4). Flowpath design studies consider the physical layout of complex flowpaths and a single loop. Fleet size studies estimate the total vehicle time needed in a shift and thus determine the number of AGVs required. Operational issues include job scheduling, AGV scheduling and dispatching, and conflict-free routing, and are classified according to the type of AGV flowpath layout. There are analytical studies for single line and single loop AGV systems, but the primary tools for complex networks of AGV paths are simulation and Petri net based simulation. When a network is considered, the problem becomes more complex because of the availability of many routes for AGV dispatching, and the possibility of conflicts between AGVs.

Several earlier reviews of the AGV systems literature have addressed one or more of the above issues, but in isolation from others. Co and Tanchoco [23] summarize the research up to 1990 on vehicle management in AGV systems. They unify various dispatching strategies, and routing and scheduling methods. King and Wilson [92] integrate the research up to 1990 on flowpath design, AGV fleet size determination, routing and scheduling methods, and the justification for and implementation of AGV systems. There has been intensive research on these issues in the last five years. Johnson and Brandeau [83] unify the stochastic models available for fleet size determination and routing of AGVs. Their coverage extends to material storage systems such as automated storage and retrieval systems. We do not address the justification and implementation issues of AGV systems since we feel that they do not fall into the domain of an OR/MS researcher or practitioner. The main intention of our work is to unify all the OR issues related to AGV systems and thus present a holistic view of the interrelations between them. As discussed below, these issues cannot be treated in isolation. As a result of this perspective, we are able to identify clearly both the gaps in the research and the new issues emerging from technological advances. We hope

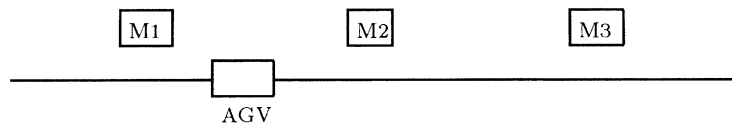


Figure 1. Single line AGV system.

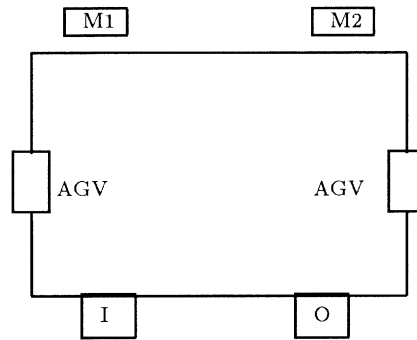


Figure 2. Single loop AGV system.

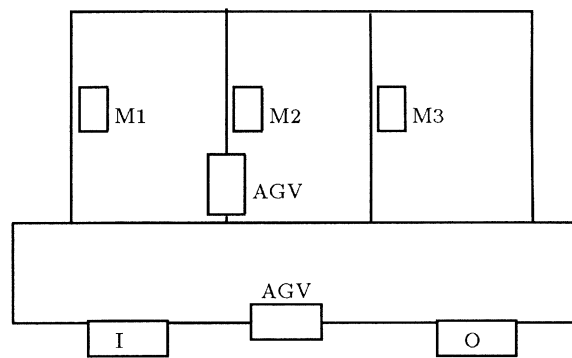


Figure 3. Ladder type AGV system.

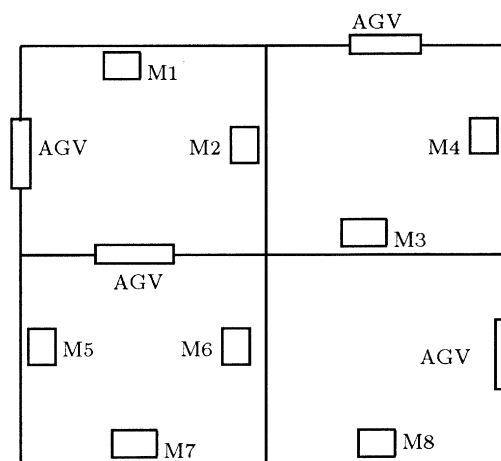


Figure 4. Complex AGV network system.

our review will be a valuable compendium to an OR/MS researcher working in AGV systems in the future.

This paper is organized as follows. The design of flowpaths is discussed in section 2.1. This section is further divided to distinguish the related studies, based on whether they assume fixed pick up and drop (P/D) points or determine the locations for them. The design of a single loop, segmented flowpaths, and virtual flowpaths are also included there. Fleet sizing issues are studied in section 2.2. Section 2.2.1 summarizes analytical studies, and section 2.2.2 does the same for simulation-based studies. Section 3.1 reviews operational issues in single line AGV systems. Section 3.2 summarizes the work on operational issues for single loop AGV systems, and section 3.3 does the same for complex AGV networks. Section 4 contains a description of several important future research issues. Section 5 provides a conclusion.

2. Design issues

In considering the design of an AGV system, we assume that the *departmental* layout is already given. The departmental layout is determined based on the volume of material flow between departments, with the objective of minimizing the total material flow. The material flow between departments in a specified planning horizon is usually represented in the form of a From-To chart. This facility layout problem is actually a quadratic assignment problem (QAP), which is unary NP-hard. See Garey and Johnson [46] for related definitions. Thus, practitioners use software packages such as CRAFT for laying out departments heuristically. In such a layout, departments are defined by rectangles whose edges are called *aisles*. Within each aisle, traffic may flow in one or more *lanes*. An excellent discussion of the facility layout problem is available in Francis et al. [41].

For a given departmental layout, the design of an AGV system should include flowpath layout, location of P/D points for each department, and the AGV fleet size. The design of the guidepath and the location of P/D points have a significant effect on the installation cost, travel time and operating expense of the system. The AGV fleet sizing problem is to determine the minimum number of AGVs required to achieve a prespecified throughput rate. Since considerable capital investment is needed for the AGV fleet, fleet sizing is an important problem. The AGV system design problem has an almost exact analog in the field of *personal rapid transit* (PRT), also called *area wide individual transit* systems design. PRT is a transportation system consisting of a network of limited-access guidepaths, conveying small passenger vehicles under automatic control. PRT is proposed and evaluated by Wolf [181], Cole [25], and Haidu et al. [63]. However, PRT does not consider individual passenger transport requests, while AGV systems do. PRT keeps on operating based on the bulk demand of the passengers. Also, a PRT vehicle can carry many passengers with several different origins and destinations, whereas AGVs usually carry one or at most a few *unit loads*, as defined in the next paragraph.

AGVs often carry a unit load. A unit load consists of one or more components kept together and transported as a unit. Each job that arrives in the shop has a certain number of identical components. The job is broken down into unit loads. Unit loads belonging to the same job follow the same job route. A job is not completed until its last unit load is delivered. Large unit loads result in the need for fewer AGVs, but require sophisticated handling, and more clamping devices and fixtures. On the other hand, small unit loads require more AGVs but reduce the cost per AGV. This highlights the importance of unit load sizing, which is discussed by Tanchoco and Agee [174], Egbelu [32, 34], Mahadevan and Narendran [120], and Hwang et al. [74]. Since unit load sizing is mainly a short-term planning issue, we do not discuss it in detail. Multiple load AGVs, however, carry two or more unit loads that may be transported to different destinations. Lastly, AGVs may be operated in unidirectional or bidirectional mode along a segment of the flowpath. However, a unidirectional mode is frequently used for easier control.

In the following two subsections, we discuss the issues involved in flowpath design (including the location of P/D points) and fleet sizing, and summarize the related literature. The underlying mathematical models are briefly discussed. We assume here that the AGVs are unidirectional and carry unit loads unless otherwise mentioned.

2.1. Flowpath design and location of P/D points

The flowpath design problem is to determine the aisles which are to be included and the direction of travel in each selected aisle, in order to minimize the total travel distance of AGVs over the planning horizon. When AGVs carry out their material transport assignments, they may have to travel empty between two P/D points. Hence, the total vehicle travel distance includes loaded travel distance and empty travel distance. Since estimating empty travel distance is complicated, as will be discussed in section 2.2, most of the studies on flowpath design consider only loaded travel distance. However, Sun and Tchernev [169] illustrate computationally that neglecting empty vehicle flow may result in a flowpath that is far from optimal.

The categorization of flowpath studies requires a brief history of the related research. Several studies assume that P/D points are given and design a complex flowpath [48, 86, 99, 156, 178]. Other studies address the problem of locating P/D points in conjunction with flowpath design, again for a complex layout [53]. Since complex layouts pose many challenges in traffic congestion and collision, still other studies address the design of a single loop that passes through all the departments [160, 176]. Later works address the design of segmented flowpaths, where flowpaths are unclosed segments that do not overlap [161, 162]. AGVs without flowpaths have also been discussed. These are called free ranging AGVs and travel along virtual flowpaths [49, 149].

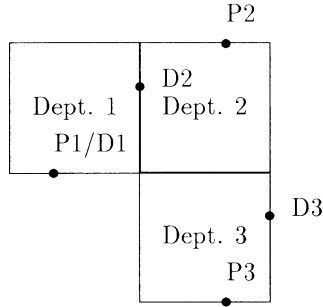


Figure 5. Example departmental layout.

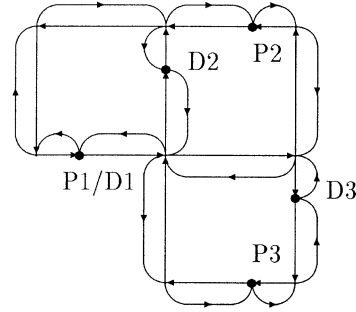


Figure 6. Planar graph for the example of figure 5.

2.1.1. Fixed P/D points

A given departmental layout can be represented as a planar graph G whose vertices, denoted by V , represent corners, P/D points, and intersections of the given layout, and whose edges represent aisles for travel between vertices. For the example departmental layout shown in figure 5, the graphical representation would be as in figure 6.

The flowpath design problem is to find a strongly connected directed subgraph G' which minimizes the total loaded travel distance of the vehicles. Let V' be the vertices of G' ($V' \subseteq V$). A succinct version of the mathematical formulation of this problem is:

$$\begin{aligned} & \text{Minimize} && \sum_{\forall(P,D)} L_{PD} * S_{PD} \\ & \text{subject to} && e_{ij} + e_{ji} \leq 1, \quad \forall(i, j) \in V, \\ & && G' \text{ is strongly connected,} \\ & && V' \text{ must include all P/D points,} \end{aligned}$$

where

L_{PD} = number of loads shipped per unit time from pick up station P to drop off station D ;

S_{PD} = shortest path from P to D , for fixed e_{ij} values;

$$e_{ij} = \begin{cases} 1 & \text{if edge } (i, j) \text{ is directed from } i \rightarrow j, \\ 0 & \text{otherwise.} \end{cases}$$

In this formulation, the first set of constraints ensures that the travel between any two adjacent vertices is unidirectional. The second and third sets of constraints ensure that a vehicle can start from any pick up point i , visit any other delivery point j , and return to i . This zero-one integer programming problem is intractable for large scale problems. A feasible flowpath for the example layout is given in figure 7.

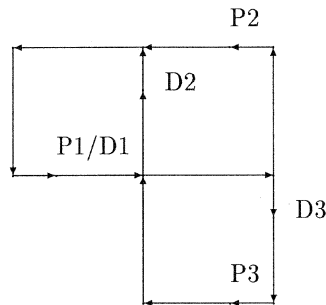


Figure 7. A feasible flowpath for the example of figure 5.

Gaskins and Tanchoco [48] consider the above problem, and solve small examples using the Multipurpose Optimization System computer package [24]. Kaspi and Tanchoco [86] describe a branch and bound approach. Sinriech and Tanchoco [156] improve that procedure by considering only the intersection nodes, which they show are sufficient. This improved method reduces the computation time by about 70% for an example with 11 departments. Venkataramanan and Wilson [178] provide a branch and bound methodology, which is similar to one proposed by Little et al. [116] for solving the traveling salesman problem [105]. The relationship between the problems is that the flowpath design problem differs from the traveling salesman problem in two ways: (1) the objective, and (2) the degree of nodes in the flowpath can be more than two. Venkataramanan and Wilson [178] extend their work to minimize the empty vehicle travel distance also. Here they distinguish two cases. In the first case, minimizing the empty vehicle travel distance is a secondary concern. In the second case, empty and loaded vehicle travel distance are equally important. Kouvelis et al. [99] present five different heuristics as well as a variety of simulated annealing approaches. Their computational study shows that a composite of these five heuristics performs better than a simulated annealing approach.

The above flowpath design formulation does not take into account empty vehicle travel and vehicle congestion. Further, it is assumed that the AGV takes the shortest path. This assumption is not valid because the AGV may take another longer path without congestion and get to the destination faster. For these two reasons, the solution found in these models is an approximate one, and it is not worthwhile to devote a lot of time to finding the exact optimal solution.

All the above models for flowpath design have an implicit assumption that an AGV remains at the point of last delivery until the next call. Another approach is to send all the AGVs to fixed terminal positions in the network. Majety and Wang [123] present a zero-one nonlinear integer programming model that simultaneously determines the flowpath and terminal locations. They linearize the nonlinear terms and solve the resulting integer programming problem using the LINDO optimizer. It takes 45 to 60 minutes on average to solve a problem instance with nine departments on an IBM mainframe computer.

2.1.2. Variable P/D points

In some design situations, options exist as to where to locate the P/D points within each department. The location of P/D points is very important as they significantly influence the traffic intensity on the aisles, the distances between departmental P/D points, and traffic control. Thus, P/D locations can affect system efficiency and consequently system cost. Goetz and Egbelu [53] study flowpath design and the location of P/D points simultaneously using integer linear programming. Even when considering only a few options for the P/D points, the problem size is already large. They propose reducing the problem size by considering only the larger flows between departments. Kim and Klein [88] consider the problem of finding P/D points for a *given* flowpath. They formulate this as a Quadratic Assignment Problem and suggest heuristics to solve it. They also extend their work to consider continuous P/D locations, where the P/D points can be located anywhere along the edges, as considered earlier by Kiran and Tansel [93].

2.1.3. Single loops

The complex AGV path found by the flowpath design models discussed above may result in AGV conflicts, at the intersections and along the aisles. One way to eliminate these conflicts is to design a unidirectional single loop that passes through all the departments. Sinriech [155], Tanchoco and Sinriech [176], and Sinriech and Tanchoco [160] study the problem of designing an optimal single loop. The single loop design problem involves, first, finding a loop that passes through all the workstations and that minimizes the total time the AGV has to travel to complete its assignments, and second, locating the P/D points for each workstation. This problem can be formulated as a large scale zero–one integer program, as in Sinriech [155]. Because such programs are hard to solve, heuristics are suggested. Tanchoco and Sinriech [176] present a three-phase solution procedure. Phase I contains an integer program to find a valid single loop, i.e., a loop that contains at least one arc of each department in the layout. Phase II includes two complete enumerations: the first begins with the valid single loop found in Phase I, and explores all possible loops that extend it, and the second eliminates loops that are dominated by others. Phase III is another MIP model which is applied to all remaining loops to find their P/D point locations. By comparing the total AGV travel distance of all remaining loops, the best single loop is selected. This work also provides a lower bound calculation procedure for a given loop, which makes Phase III work faster. This lower bound procedure calculates the lowest possible travel distance required between departments which do not share any segment of the loop, by locating the P/D points in the best possible way.

Sinriech and Tanchoco [160] extend the above work by providing a heuristic for obtaining a valid single loop, and a modified approach for Phase III, which exploits characteristics of the From-To chart. Sinriech and Tanchoco [158] locate P/D points in a given single loop containing both inter- and intra-departmental flows, by using a mixed integer program. This is the first work to consider intra-departmental flows.

Sinriech and Tanchoco [159] demonstrate computationally that single loop design is not very much affected by neglecting empty vehicle travel. Asef-Vaziri and Sriskandarajah [5] develop an integer program to integrate the design of a single loop and the location of P/D points with various objectives such as minimizing total loop length, minimizing loaded travel time, and minimizing total installation and operation cost. They use the mixed integer optimization code CPLEX 3.0 to solve practical size problems with this formulation. Exact solutions are found for problems with as many as 11 departments in less than 9 minutes on a SUN SPARC Station 20 Model 50. They also extend their work to determine whether locating multiple P/D points within a department, and whether operating AGVs in bidirectional mode, are economical.

2.1.4. Unidirectional versus bidirectional paths

We reiterate that most of the above studies consider only the design of unidirectional flowpaths. The main reasons for the use of unidirectional networks are simplicity in design and control. Encouraged by similar studies for railroad traffic management (for example, Frank [42] and Peterson [141]), Egbelu and Tanchoco [38] show by simulation that a bidirectional flowpath has performance advantages over traditional unidirectional configurations in some AGV systems, especially those that require fewer vehicles. The performance measure considered is the number of unit loads completed over a given period of time. Kim and Tanchoco [91] use simulation to compare unidirectional and bidirectional AGV systems and arrive at the same conclusions. They consider throughput rate and mean flow time (or average completion time) of jobs objectives for comparing the unidirectional and bidirectional paths. They also indicate that bidirectional paths result in increased network reliability due to the ability to reroute the AGV in case a path is blocked.

2.1.5. Segmented flow paths

Some of the goals of simplicity, efficiency and cost effectiveness are met by using a single loop. However, the performance of these flowpaths is impaired by the physical constraints imposed, such as limiting the flowpath to a loop or limiting the number of AGVs to one per loop. In order to solve some of these problems, Sinriech and Tanchoco [161] and Sinriech et al. [162] propose a segmented flow approach. The segmented flowpath is comprised of one or more zones, each of which is separated into nonoverlapping segments served by a single AGV (figure 8). Transfer buffers are located at both ends of each segment and serve as interface devices between the segments, where an AGV can deposit loads which are destined for other segments and pick up loads that have arrived from other segments. The AGV is *bidirectional* on each segment.

Sinriech and Tanchoco [161] provide a mixed 0–1 integer programming model for the design of segmented flowpaths. The objective function minimizes the cost of setting up P/D stations and the transportation cost. They consider multiple P/D

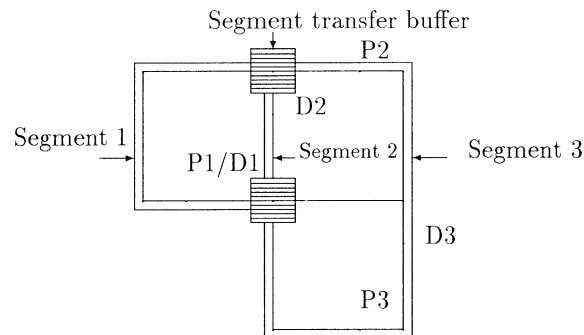


Figure 8. A 3-segmented flowpath for the example of figure 5.

points, which may reduce the transportation cost drastically. They solve a problem with 11 departments using the CPLEX mixed integer optimizer in 1,860 minutes on a GOULD NP1 computer. Sinriech et al. [162] propose segmenting a single loop and operating one AGV in each segment in bidirectional mode. The initial optimal single loop is obtained as in Tanchoco and Sinriech [176], discussed above. The segmentation is carried out using a heuristic, and the solution is improved through simulation. The primary objective is to minimize the segment to segment transfer, and the secondary objective is to balance the workload among all segments.

2.1.6. Virtual flow paths

The above flowpath design models use a From-To chart to estimate the total loaded travel distance. In a typical modern manufacturing system, the From-To chart changes over time when the part mix changes. If a flowpath has been designed using a From-To chart, that flowpath may no longer be optimal when the From-To chart changes. This illustrates the inflexibility of physically guided AGVs. In order to avoid this inflexibility, researchers are developing *free ranging* AGV systems, which do not use a physical guidepath and hence are also called *virtual path* AGV systems. The flowpath of these systems exists only in computer memory. Broadbent et al. [18] discuss the technical details, advantages and future research issues of free ranging AGVs, using their experience with the prototype built at Imperial College, London. Similar studies on free ranging AGVs are underway at Purdue University [89]. Katz and Asbury [87] discuss the hardware and software requirements for navigation of such AGVs. Gaskins et al. [49] study the virtual flowpath design problem for free ranging AGVs. Both the number of virtual lanes of vehicle flow in each aisle and the direction of flow in each lane must be determined. A multi-commodity network flow model is presented, considering loaded vehicle flow and empty vehicle flow. The objective is to minimize the sum of total vehicle travel distance and the total number of lanes. The main limitations of the model are the computational difficulty encountered, and not considering vehicle conflicts.

Seo and Egbelu [149] also provide a model to design flexible guide paths. They formulate the problem as a mixed integer program and suggest a two-phase heuristic. Phase I minimizes the loaded vehicle flow. Since only the loaded flow is considered in Phase I, the solution obtained in Phase I may be incomplete for empty vehicle travel, i.e., there may not be any path available for empty vehicle travel between two departments. In such cases, Phase II redesigns the flowpath, taking into account empty vehicle travel distance.

2.2. Fleet sizing

The determination of the number of AGVs needed for the system to provide a given throughput rate is the next step after the flowpath has been laid out. The number of AGVs required, or fleet size, can be found if the total vehicle travel time required in the planning horizon is known. The total vehicle travel time divided by the length of the planning horizon is a lower bound on the required fleet size. Total vehicle travel time includes loaded travel time, empty travel time, and time spent in loading and unloading. The loaded travel time can be found approximately using the From-To chart, assuming that AGVs take the shortest path to carry out their assignments. In reality, an AGV may not always take the shortest path, due to conflict with other AGVs. The optimal way to route AGVs, taking into account conflicts, is discussed in section 3.3.1. The mathematical model presented in section 2.1.1 estimates a lower bound on the loaded vehicle travel distance, since it uses shortest paths and ignores AGV conflicts. This lower bound is a useful starting point for any detailed fleet size estimation study. The estimation of empty vehicle travel time is a complex task, since it requires information about how the empty vehicles are dispatched. Various dispatching policies are discussed in section 3.3.3. The time spent in loading and unloading has to be estimated from the number of loadings and unloadings performed in a planning horizon or shift. Thus, analytical studies can only roughly estimate the fleet size. Due to the complex nature of the issues involved in determining fleet size, simulation seems to be the most promising tool. However, the fleet size estimate obtained from an analytical model can be a good starting point for a detailed simulation. We summarize the various analytical methods in section 2.2.1. We outline the issues considered in simulation and other studies in section 2.2.2.

2.2.1. Analytical studies

An AGV-served manufacturing system, as shown in figure 9, is discussed in Bozer et al. [13]. In figure 9, the input/output (I/O) stations are represented by stations 1 and 2, and the processor stations are denoted by stations 3 and 4. There is an input queue and an output queue of sufficient capacity for each station, including the I/O stations. Jobs from outside the system enter through one of the I/O stations and, when all the operations have been completed, they exit through one of the I/O stations. An incoming job arrives directly at the input queue of an I/O station, while an outgoing

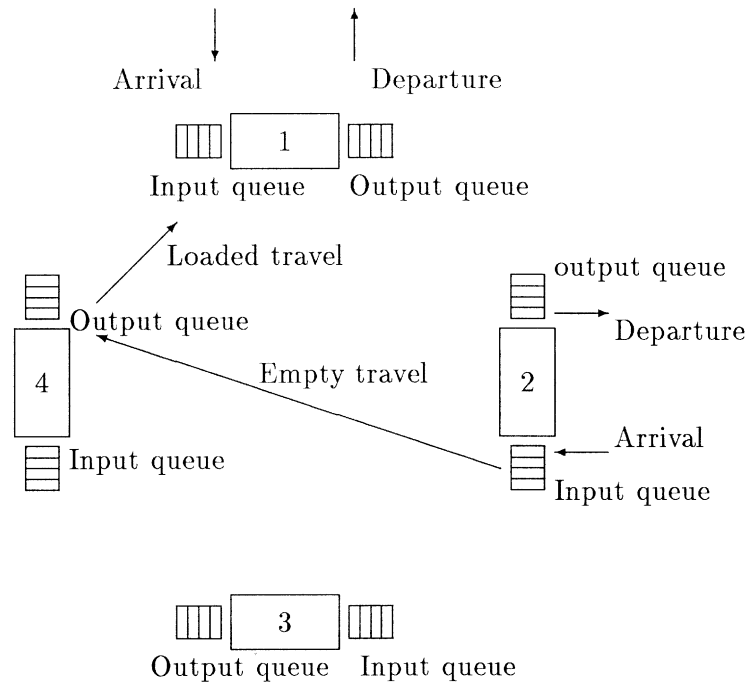


Figure 9. AGV-served manufacturing system.

job is deposited at the output queue of an I/O station, where it is assumed to exit from the system instantly. That is, no processing takes place at the I/O stations. The work on fleet size estimation can be further classified into deterministic and stochastic studies. If the trip demand is known in advance, then we have a deterministic system. Otherwise, stochastic models are needed to estimate the empty vehicle travel. We now summarize the studies on deterministic systems followed by those on stochastic systems.

Deterministic demand

The seminal work on this topic is the analytical model of Maxwell and Muckstadt [128]. The net flow of vehicles in each department during a shift or planning horizon is the difference between the numbers of incoming and outgoing vehicles, and can be estimated using the From-To chart. Typically, some departments will have positive net flow and others will have zero or negative net flow. The flow balance of departments has to be achieved by empty vehicle movement. To minimize the total empty vehicle travel time, the vehicles from the departments with positive net flow are optimally dispatched to the departments with negative net flow. This is a classical transportation problem which can be solved efficiently. To account for time-dependent effects such as AGV conflicts, the authors assume an empty vehicle dispatching policy

that balances the time between AGV visits at each department. A deterministic simulation estimates the delay due to conflict, and also shows that the buffer size at P/D points significantly influences the amount of delay due to AGV conflicts. Malmborg [125] presents an improved lower bound on the empty vehicle travel time using the concept of expectation in probability theory. By explicitly considering the dispatching policy in use, this model estimates the expected empty travel time. Malmborg and Shen [126] extend this model to find the fleet size. They compare analytical and simulation models for two different AGV dispatching policies. They find that the analytical model deviates significantly from the simulation model only when the fleet size is high. Even in this case, they conclude that the analytical model provides a good starting point for a detailed simulation.

Egbelu [31] presents four analytical models to estimate fleet size. The first model assumes that the amount of empty vehicle travel is equal to the amount of loaded vehicle travel. The second model estimates the delay time due to AGV conflicts and vehicle idle time by multiplying the loaded travel time by constants, but does not account for empty vehicle time. The third model accounts for inter- and intra-departmental empty vehicle time. The net flow of vehicles at each department is calculated as in Maxwell and Muckstadt [128]. The total number of empty trips is the sum of all positive net flows or the sum of all negative net flows, whichever is smaller. The author estimates the empty travel time by multiplying the total number of empty trips by the average time taken for a loaded trip. The empty intra-departmental travel time is estimated by multiplying the minimum of the number of incoming and outgoing flows at that department by the travel time between the pick-up and delivery points. The fourth model estimates the empty vehicle time using expectations. Here, the expected number of empty trips from department i to department j is the product of the expected total pick-ups made at j and the expected total drops made at i . He uses the simulation package AGVSim [36] to estimate the exact number of AGVs required, assuming the AGV dispatching policies discussed in Egbelu and Tanchoco [37], and finds that all four analytical models provide good starting points for detailed simulation studies.

Stochastic demand

Johnson [79] explains how to estimate the fleet size in a system where the jobs arrive according to a Poisson process. He assumes a First-Come-First-Served (FCFS) rule for empty vehicle dispatching. Let λ_{ij} be the average demand rate for loaded trips from station i to station j . At any time, the probability that the next trip is from i to j is

$$w_{ij} = \lambda_{ij} / \Lambda,$$

where $\Lambda = \sum_i \sum_j \lambda_{ij}$. Let $q_{i|j}$ be the probability that a vehicle comes from station i , given that a call for transport originates from station j . This is the probability that a vehicle is waiting at station i , and is given by

$$q_{i|j} = \sum_k w_{ki}.$$

If we define v_j as the probability that the next transport request comes from station j , then

$$v_j = \sum_h w_{jh}.$$

The probability that the next trip is an empty trip from i to j is the probability that the last order was dropped off at station i and the next order must be picked up at node j , which is given by

$$q_{ij|FCFS} = \sum_k w_{ki} \sum_h w_{jh} = q_{i|j} v_j.$$

The expected number of trips from i to j is $q_{ij} \Lambda$ and the expected length of an empty trip is given by

$$E(\text{empty} | FCFS) = \sum_i \sum_j q_{ij} d_{ij},$$

where d_{ij} is the shortest travel time from i to j (a function of the distance and velocity of the vehicle). One can approximate the expected time for which a job waits for transportation using an $M/G/c$ queueing model with general service times and c vehicles. To estimate the expected waiting time in the system, the author uses an expression originally derived by Hokstad [72]:

$$E(\text{wait}) = \frac{1}{2} \frac{E(s^2)}{(E(s))^2} \frac{(c\rho)^c}{(c-1)! E(s)(c/E(s) - \Lambda)^2} \mathcal{P}_c,$$

where $\mathcal{P}_c$ is the probability that all c vehicles are idle as detailed in the cited work, $\rho = \Lambda E(s)/c$ is the vehicle utilization, and $E(s)$ is the expected total trip time. The expected total trip time includes expected loaded travel, expected empty travel, loading time (l), and unloading time (u), and hence

$$E(s) = \sum_i \sum_j w_{ij} d_{ij} + \sum_i \sum_j q_{ij} d_{ij} + l + u.$$

The required fleet size, c , can be determined by setting $E(\text{wait})$ to a value desired by management.

Johnson [79] also develops a model similar to the one above under the Nearest Vehicle (NV) AGV dispatching policy. Bozer et al. [13] present a queueing model to estimate the expected time for which a job waits for transportation in a manufacturing system with one AGV, under the assumption that empty vehicles are dispatched according to a modified FCFS policy. This model can be extended to find the fleet size, as above. Other analytical studies are summarized in table 1. The second column

Table 1

Analytical fleet sizing studies.

Author(s)	Deterministic/ stochastic	OR model	Empty travel	Vehicle congestion	Main conclusions or special issues
Maxwell and Muckstadt [128]	D	Transportation model	Yes	Yes	Finds the model useful in real systems design they undertake.
Egbelu [31]	D	Model 1	Yes	No	All four models are good starting points for simulation.
	D	Model 2	No	Yes	
	D	Model 3	Yes	No	
	D	Model 4	Yes	No	
Tanchoco et al. [175]	S	CAN-Q	No	No	Good starting point for simulation.
Wysk et al. [182]	S	CAN-Q and spreadsheet	No	No	Spreadsheet makes iterative design faster.
Leung et al. [110]	D	Mixed integer programming	Yes	No	Multiple load carrying AGVs are considered.
Jafari [76]	S	Queueing theory	No	No	Validates with SIMAN.
Malmborg [124]	S	Queueing theory	Yes	Yes	Variation in fleet size can significantly affect shop locking (see section 3.3.3).
Malmborg [125]	S	Queueing theory	Yes	Yes	Better empty travel than earlier studies.
Mahadevan and Narendran [119]	D	Expectation	No	No	Routing flexibility of FMS is considered.
Sinriech and Tanchoco [157]	S	Goal programming. Maximizes throughput, minimizes cost	No	No	Adding AGVs after a point decreases through- put due to congestion.
Egbelu [34]	D	Mixed integer programming	No	No	Integrate with unit load sizing and vehicle size selection.
Mahadevan and Narendran [121]	S	Probability	Yes	No	Machine failure and routing flexibility are considered. Simulation validates the model.
Mahadevan and Narendran [122]	S	Probability	Yes	No	Same as above and also single vehicle and multi- vehicle loops are com- pared. Single vehicle loops are less tolerant for machine failures.
Srinivasan et al. [169]	S	Queueing theory	Yes	No	Explicit consideration of AGV dispatching policy (MFCFS).
Malmborg and Shen [126]	S	Queueing theory	Yes	No	Simulation validates the model.

in table 1 indicates whether the study considers a deterministic job set or stochastically arriving jobs. The third column shows the main modeling tool used. The abbreviation CAN-Q stands for computer analysis of networks of queues, and this software is described by Solberg [163]. The fourth and fifth columns show whether empty travel and vehicle congestion, respectively, are considered in the study.

The above analytical fleet sizing studies for AGV can be complemented with studies for crane-served manufacturing systems. For example, Lei et al. [108] and Armstrong et al. [4] provide fleet sizing algorithms for *single-track* crane-served manufacturing systems. They consider a crane-served electroplating system, where the processing time of a job at a stage must be within given lower and upper limits, and the jobs are identical. Their results also hold for a manufacturing system that is served by AGVs in bidirectional mode.

2.2.2. Simulation, expert systems, and Petri net methods

Table 2 summarizes the simulation studies on AGV fleet sizing. The simulation language used, AGV dispatching policy assumed, and special issues are listed. The third column abbreviates the AGV dispatching policies, as described in table 3 in

Table 2
Simulation-based fleet sizing studies.

Author(s)	Language	AGV dispatching policy	Main conclusions or special issues
Newton [134]	Fortran	VLFW	Compares VLFW with FIFO.
Egbelu [31]	AGVSim	All policies in Egbelu and Tanchoco [37], see table 4	Studies effect of flow volume, dispatching policy and layout on fleet size. Dispatching policy significantly influences the fleet size.
Ozden [138]	LISP	FEFS	Multiple load AGVs.
Lee et al. [107]	SIMAN	Nearest vehicle	Exponential job arrival has higher throughput than uniform arrivals.
Gobal and Kasilingam [52]	SIMAN	STT	Objective is to balance the job, machine and AGV idle times.
Occeña and Yokota [136]	Not available	Maximum demand	JIT system is considered.
McHaney [129]	GPSS/H	STT	Battery constraints also modeled.
Shang [151]	SLAM II	FCFS, STT, LQS	Taguchi method is used for robust design.

section 3.3.3. The simulation languages are capable of modeling empty vehicle travel and vehicle congestion, and therefore all the studies on simulation account for both empty vehicle travel and vehicle congestion.

Grosseschallau and Kusiak [62] describe an expert system, INSIMAS, which can carry out the detailed design of an AGV system, including the fleet size. INSIMAS divides the design process into four phases: modeling, analysis, simulation, and evaluation. Raju and Chetty [142] use a Petri net based methodology for fleet sizing. They introduce an extended timed Petri net to model the dynamics of AGV systems. They develop an Ada based software package, SIMAGVS, in order to implement their Petri net. Using examples, they illustrate the convenience and efficiency of SIMAGVS.

2.2.3. Differing environments

Jaikumar and Solomon [77] consider the fleet sizing problem integrated with machine scheduling and AGV scheduling in a manufacturing system where jobs are returned to the central warehouse after each machining step. We discuss this work further in section 3.3.2. The design issues that arise during the replacement of a non-automated material handling system by an automated one such as an AGV system are also interesting. During the transition period, managers want to know how many AGVs should replace existing devices and which workstations should be serviced by the automated material handling system. Other workstations will continue to be served by nonautomated devices. Such design problems usually arise in “clean” environments such as electronics manufacturing, where AGVs are used to replace nonautomated devices which transport production components from the central warehouse to the workstations. Other material transport within the system is handled differently.

Johnson and Brandeau [81] consider the introduction of a single AGV into such systems and address the problem of which workstations should be served by this AGV. A queueing model estimates the expected waiting time of jobs in the system for material transport. This model explicitly considers the empty travel time of the AGV. The job waiting time is embedded into a zero–one integer program to determine the workstations that should be serviced by the AGV. The objective is to maximize the net benefit of the AGV system, which is measured by net present value of labor savings minus the cost of constructing and operating the AGV system. One constraint of the zero–one integer program ensures that the expected waiting time of a job for material transport does not exceed a prespecified value set by management. The authors also provide an implicit enumeration procedure to solve the zero–one integer program. Johnson and Brandeau [80] consider how to find the minimum number of AGVs that are needed to replace nonautomated devices. This model also determines which workstations should be serviced by those AGVs. Johnson and Brandeau [82] extend their 1993 work to consider AGVs with multiple load carrying capacity.

3. Operational issues

Studies on operational issues assume that the AGV flowpath is given, and that the fleet size has also been determined. The operational issues arising in an AGV-served manufacturing system vary significantly depending upon the layout of the

system. In a single line layout such as an assembly line, it is necessary to schedule the jobs and determine the AGV launch times. In loop layouts, job scheduling methods, AGV operation policy and conflict avoidance methods need to be specified. Conflict avoidance is usually achieved by a *zone control* approach, where the flowpath is divided into many invisible zones, and only one AGV is allowed in a zone at any time. Another method of avoiding conflict is called forward sensing: the vehicle senses when another vehicle is close to it and stops. Even though these technologies exist to prevent conflicts, they may still occur. An important task of an OR/MS researcher is preventing the occurrence of such AGV conflicts, which task is called conflict-free routing. In a more complex layout, job scheduling, AGV dispatching and conflict-free routing need to be addressed. In the following three subsections, we describe the operational issues and unify the studies undertaken in these layout configurations. We also show how operational issues in other environments relate to those for AGV systems.

3.1. Single line layout

A single line layout is illustrated in figure 1. Johnson [84] proposes a polynomial time algorithm for a two-machine flowshop scheduling problem with the objective of minimizing the completion time of a set of jobs, or makespan. In a flowshop, each job is processed on the same sequence of machines. This study assumes that the time taken to transfer a finished job from the first machine to the second machine is negligible, and that a buffer which can hold all the jobs is available between the machines. A natural extension of this work is to consider material handling time also. Maggu et al. [118], Stern and Vitner [168] and Panwalkar [139] consider makespan minimization problems where AGVs are used to transfer the finished job from the first machine to the second machine. Park and Posner [140] consider similar problems with crane transporters. The number of AGVs used for material handling, and the availability of temporary storage between the machines, determine the complexity of the scheduling problem. When the number of AGVs is unlimited and there is an infinite capacity buffer between the machines, a finished job on the first machine can immediately be transported and placed in that buffer. As shown by Maggu et al. [118], by adding the transportation time of each job to the processing time on both machines, this problem reduces to that of Johnson [84]. Stern and Vitner [168] show that, when there is no buffer between the machines, only one transporter, and the transportation time between machines is job dependent, this problem can be formulated as an asymmetric traveling salesman problem. The details of their formulation are as follows. Let a_j and b_j denote the processing times of job j on machines 1 and 2, respectively. Let t_j denote the transportation time of job j from machine 1 to machine 2. The time for an empty AGV to return from machine 2 to machine 1 is denoted by r . Let $t'_j = t_j + r$. Finally, let $\sigma(k)$ denote the job scheduled in k th position in sequence σ . Then the makespan, $Z(\sigma)$, for the production of n jobs in sequence σ is given by Stern and Vitner [168] as

$$\begin{aligned}
Z(\sigma) &= a_{\sigma(1)} + t_{\sigma(1)} + \sum_{k=1}^n b_{\sigma(k)} \\
&\quad + \sum_{k=2}^n \max\{0, r + t_{\sigma(k)} - b_{\sigma(k-1)}, a_{\sigma(k)} + t_{\sigma(k)} - t_{\sigma(k-1)} - b_{\sigma(k-1)}\} \\
&= a_{\sigma(1)} + b_{\sigma(n)} - r + \sum_{k=1}^n t'_k + \sum_{k=2}^n \max\{b_{\sigma(k-1)} - t'_{\sigma(k)}, 0, a_{\sigma(k)} - t'_{\sigma(k-1)}\}.
\end{aligned}$$

We now show that this makespan minimization problem is intractable.

Lemma 1. The scheduling problem considered by Stern and Vitner [168] is unary NP-hard.

Proof. By reduction from the classical scheduling problem $F3|no-wait|C_{\max}$, as defined by the notation of Graham et al. [60]. Röck [144] shows that this makespan minimization problem in a three-machine no-wait flowshop is unary NP-hard. In a no-wait flowshop, each job must be processed without interruption from its start on the first machine to its completion on the last machine [67]. Let x_j, y_j and z_j denote the processing time of job j on the first, second and third machine, respectively, in the three-machine no-wait flowshop. Then the makespan, $C(\sigma)$, for the production of n jobs in sequence σ can be written (see Hall and Sriskandarajah [67] for details) as

$$\begin{aligned}
C(\sigma) &= x_{\sigma(1)} + y_{\sigma(1)} + z_{\sigma(1)} \\
&\quad + \sum_{k=2}^n \max\{z_{\sigma(k)}, z_{\sigma(k)} + y_{\sigma(k)} - z_{\sigma(k-1)}, z_{\sigma(k)} + y_{\sigma(k)} + x_{\sigma(k)} - z_{\sigma(k-1)} - y_{\sigma(k-1)}\} \\
&= x_{\sigma(1)} + z_{\sigma(n)} + \sum_{k=1}^n y_k + \sum_{k=2}^n \max\{z_{\sigma(k-1)} - y_{\sigma(k)}, 0, x_{\sigma(k)} - y_{\sigma(k-1)}\}.
\end{aligned}$$

To construct an instance of the problem considered by Stern and Vitner [168], we set $a_i = x_i$, $b_i = z_i$ and $t'_i = y_i$. It is immediately clear that minimizing $C(\sigma)$ is equivalent to minimizing $Z(\sigma)$. $\square$

Panwalkar [139] provides a polynomial time algorithm for the case of one AGV and an infinite buffer. Levner et al. [111] generalize this result to a similar problem with job-dependent transportation times and non-negligible load and unload times. Kise [94] considers a different two-machine flowshop model where each machine has an unlimited intermediate buffer capacity, and a single AGV transports semi-finished jobs deposited at the buffer of machine 1 to the buffer of machine 2. He shows that makespan minimization in this context is a binary NP-hard problem, by contrast with

the case considered by Panwalkar [139]. He also identifies polynomially solvable special cases of this problem.

Real-world flowshops, such as automobile assembly lines, often contain a large number of machines. In such assembly lines, AGVs are often used as transporters or assembly tables and dedicated to the line. The sequencing of jobs to minimize makespan in a flowshop with three or more machines is well known to be unary NP-hard [47]. The addition of material handling issues makes the problem even less tractable. Thus, much of the literature focuses on the design of good heuristics. It is also important to minimize the time that AGVs spend waiting in line, because this ties up an expensive resource. The total time an AGV spends in the line is defined as the AGV utilization. The need to minimize AGV utilization can be explained as follows. If there is a way to reduce the time an AGV spends in the line, subject to the same makespan, then the AGV time saved can be used to charge the AGV batteries and service the AGV. Alternatively, in that saved time, the AGV can be used for other material handling needs of the factory. Minimizing the makespan and minimizing the AGV utilization are conflicting objectives. For a given job sequence, if we are attempting to minimize the makespan, then AGVs should be launched as early as possible. However, when we launch AGVs as early as possible, they may have to wait in front of the machines for a longer time, thus increasing AGV utilization. To address both objectives, the line idleness and the vehicle idleness should be balanced. In a similar environment, Egbelu and Roy [35] estimate the launching times of AGVs that achieve such a balance. Grasso [61] proposes a heuristic for job sequencing based on the work of Dannenbring [26] for large flowshops. He compares three launching policies: early start, late start and no-wait. In the early start policy, AGVs are launched as soon as they are available. In the late start policy, AGVs are launched at the latest possible time such that no machine has to wait for a job. The no-wait policy ensures that AGVs do not idle in front of machines while a previous job completes processing. In computational testing with randomly generated data, the late start policy performs best for a variety of data sets using minimization of makespan and of AGV utilization as criteria.

Where the AGVs are dedicated to the line, the principal objective in scheduling becomes throughput rate maximization. Han and McGinnis [69] study a scheduling problem in a large shop where machines are arranged in a line and served by a single dedicated transporter, such as an AGV. This transporter scheduling problem is shown to be equivalent to a one machine (i.e., one AGV) due date problem with sequence dependent processing times and due dates. The complexity of this problem remains unknown, but the authors propose a heuristic in which the next AGV move is to the station with highest priority. A priority index for a workstation i is found using the travel time to i from the current transporter position and the remaining time until the due date at i . The job sequencing issue is not addressed.

It is important to understand the relationships between scheduling problems arising in different material handling environments, since this understanding provides

insights and also useful results in some cases. The mode of operation of a bidirectional AGV and that of a crane are similar from a scheduling perspective. Therefore, the results of the above studies are directly applicable to crane scheduling. Also, the results of the single crane two-machine problem with the objective of minimizing the makespan, as studied by Park and Posner [140], can be used for the single AGV two-machine makespan minimization problem. Similar comments apply to the crane scheduling study of Matsuo et al. [127]. Conversely, the results of the multiple AGVs, multiple-machine makespan minimization studies of Grasso [61] and Egbelu and Roy [35], are directly transferable to crane scheduling.

3.2. Single loop layout

A single loop layout is illustrated in figure 2. In AGV-served manufacturing systems, an AGV picks up the raw material from the input point, carries it to the first machine and loads it there. Similarly, the same device unloads the finished job from the second machine and carries it to the output point to drop it off. After dropping off the finished job, the AGV has to come back to the input point to pick up a new job. In systems that employ a unidirectional AGV, this kind of material handling can most easily be achieved by operating an AGV in a single loop. An application of a loop layout can be found at Ford Motor Company's Kansas City plant, where a fleet of 18 AGVs carries instrument panels around a 600-foot wire flowpath and through 20 manual and two automated workstations [131].

A factor that influences scheduling is the availability of buffer space to store work-in-process (WIP). Buffer storage for WIP requires large floor space and elaborate material handling equipment when jobs are big and heavy, for example engine casings of ships. Assume, for example, that AGVs are operated under a policy where they remain stationary only if they cannot do any other material handling in the loop. Kise et al. [95] consider a two-machine flowshop loop with no buffer for work-in-process, where AGVs are operated according to this policy. They provide an $O(n^3)$ time algorithm, based on the algorithm of Gilmore and Gomory [50], to find a job sequence that minimizes the makespan. We recall the work of Stern and Vitner [168] discussed in section 3.1. If the transportation times between the machines are job independent, the problem they discuss and that of Kise et al. [95] are equivalent, and hence their problem is solvable by the $O(n^3)$ time algorithm mentioned above. Levner et al. [112] generalize the work of Kise et al. [95] by providing an $O(n^3)$ algorithm for job-dependent load and unload times and machine setup times. Agnetis et al. [2] consider an AGV-served two-machine flowshop loop similar to that of Kise et al. [95], except that the machines are equipped with a swapping device, which enables an AGV that reaches a machine loaded with a job to exchange it with another job released by that machine. They study the scheduling of batches of identical jobs with the objective of minimizing the makespan, and prove that when the batch size is larger than a critical value, this scheduling problem is a polynomially solvable case of the traveling sales-

man problem. Ganesharajah et al. [45] study the cyclic scheduling of a minimal job set in a flowshop loop served by one or more AGVs. They consider three different AGV dispatching policies: early start, late start and no-wait [61]. They show that scheduling to minimize cycle time under a no-wait policy with one AGV is a polynomially solvable problem. However, they also show that, when other dispatching policies are used, or when the number of AGVs is increased, the scheduling problem is intractable.

We now relate the AGV scheduling problem to the *robotic cell* scheduling problem [65, 150]. A robotic cell consists of one robot serving two or more machines that are usually arranged in a semi-circle. There is no buffer at any machine. Sethi et al. [150] identify $m!$ potentially optimal robot routing policies in m machine robotic cells. A specific robot routing policy for $m = 2$, studied in their work, is also considered for the control of a unidirectional AGV system by Kise et al. [95]. If a bidirectional AGV is employed in a line layout, the results obtained for robotic cells are directly applicable to the case of AGVs. For example, an m machine ($m \geq 2$), single line, bidirectional single AGV problem is mathematically equivalent to the m machine robotic cell problem studied by Logendran and Sriskandarajah [117], Sriskandarajah et al. [166] and Hall et al. [65, 66]. Kamoun et al. [85] study design and operational issues in a linked-cell manufacturing system consisting of cells connected by intermediate buffers, where each cell is served by a single robot. Their results are also valid if each cell is served by an AGV in bidirectional mode in a line layout.

An FMS consists of several machine centers, linked by a central computer, which are capable of carrying out many different manufacturing operations. Thus, FMSs frequently contain identical parallel machines. Blazewicz et al. [12] study how to schedule jobs and AGVs to minimize the makespan when an AGV serves such machines in a single loop. For a given production schedule, whether the AGVs can meet job deadlines can be checked in polynomial time. If a feasible solution exists, it can be found in $O(n \log n)$ time by the minimum slack time rule: a job with minimum slack time is loaded on the vehicle first. Slack time is the remaining time before the due date. A pseudo-polynomial time algorithm, based on dynamic programming, is proposed for the simultaneous construction of optimal production and vehicle schedules. Blazewicz et al. [11] extend this work to an FMS where there are two loops for the AGV path, as shown in figure 10. All the machines in the FMS are identical parallel machines, but they are grouped into M_1 in one loop and M_2 in the other. In this case, conflicts between AGVs should also be considered. One conflict-free strategy is to make the release times of AGVs pairwise relatively prime integers, modulo the greatest common divisor of the cycle times of the two loops. The cycle time of a loop is the time taken by an AGV to traverse the complete loop. If this condition is satisfied, the AGVs will reach the common point for inspection at least one time unit apart, thus avoiding a conflict. Also, catching up conflicts are avoided by maintaining at least one time unit difference between successive dispatches. Blazewicz et al. [11] also provide a dynamic programming based approach which checks, for a given production

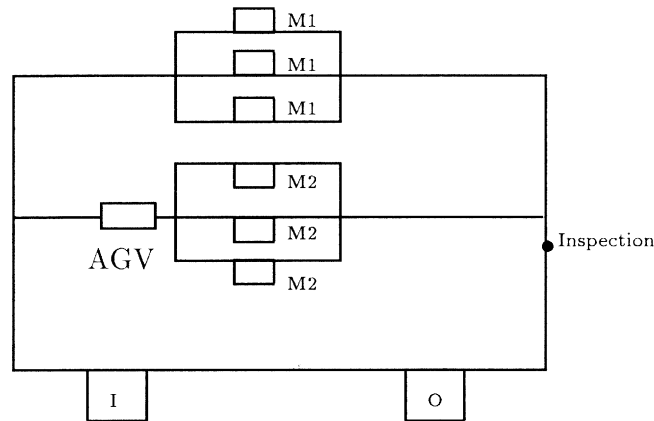


Figure 10. A two-loop FMS.

schedule, whether or not a required raw material can be delivered on time to the appropriate machine. The problem of jointly optimizing the job and AGV schedules is not addressed.

The preceding studies all relate to *static* manufacturing systems. In static systems, the job set to be processed by the system is known in advance. But in most manufacturing environments, jobs may arrive randomly. Such environments are called *dynamic* manufacturing systems. In dynamic systems, we cannot preplan the job and AGV schedules. Therefore, an important task is developing effective AGV dispatching policies. Many of these are described in section 3.3.3, but we discuss here the work of Bartholdi and Platzman [8], who consider a dynamic manufacturing system served by an AGV in a single loop. There are many machines in the system and there can be many P/D points in the loop. The problem is how to operate the AGV in the loop, i.e., in which sequence it should carry out the requests for material transfer, to minimize the average waiting time of a job. They consider a centralized greedy policy First-Come-First-Served (FCFS) and a decentralized greedy policy First-Encountered-First-Served (FEFS). In FCFS, AGVs pick up and deliver loads in the sequence in which they have arrived. In FEFS, an AGV continually circulates the loop and, whenever it has available space, it picks up the first load it encounters. Loads that are being carried are delivered whenever their destinations are encountered. They study the long-run performance of the FEFS policy by an analysis similar to that of nondeterministic single-server polling systems [173], which indicates that FEFS achieves better long-run service rates than FCFS. A simulation study confirms that FEFS outperforms FCFS in the long run. When a continuously circulating policy is not adopted, where to position an idle AGV after a delivery becomes an important consideration. The objective here is to minimize the empty vehicle travel time and thus also the system's mean response time. This problem is very similar to that of prepositioning a taxi or an ambulance for quick response. Egbelu [33] transforms a loop layout into an

approximately circular layout and provides heuristics to determine home positions in single-vehicle and multi-vehicle cases, and in unidirectional and bidirectional cases, based on the assumption that move requests at a work station are static. Chang and Egbelu [19] extend this work to consider dynamically changing move requests at a work station.

3.3. Complex network layout

Many practical AGV networks are complex. In such systems, there are many paths available for a load to be transported to its destination, as in figures 3 and 4. When several AGVs travel along different paths, the chances for AGV conflicts increase. This leads to the issue of AGV conflict avoidance. In static systems where the job set to be produced is known in advance, it is necessary to coordinate job and AGV schedules if the system is to operate close to a mathematical optimum. This leads to some complex issues of joint optimization. Also, many real systems are dynamic in the sense that the set of jobs is not known in advance, and thus it is not possible to optimize schedules, so heuristic dispatching rules are needed. The following sub-sections discuss these three issues in turn.

3.3.1. AGV conflict-free routing

Case 1: Static job set. We consider a complex network and static job set first. We have a job set to be produced using various machines. These jobs are to be transported by several AGVs between various origins and destinations. The AGV fleet size is given. The routing of AGVs without conflict is the subject of this section. A natural objective is to minimize the overall completion time, or makespan.

Conflict-free routing of AGVs is a special case of the extensively studied *vehicle routing problem* (VRP). The solution of the VRP aims at serving each customer with known demand exactly once by a fixed fleet of vehicles, so as to minimize the total route length. All vehicles are located at a central depot and each vehicle returns to the central depot after a tour. An excellent summary of the heuristics developed for the VRP is given by Golden and Assad [55].

Vehicle routing with time windows is another related class of problems. In a large number of vehicle scheduling problems, collections and deliveries are required to be made within given time windows. Examples include school busing, scheduling of urban transit, passenger trains, and airline fleets. Kolen et al. [98] and Solomon [164], among others, provide algorithms for the VRP with time windows. The class of vehicle routing problems which most resembles the AGV routing problem is dial-a-ride models [167]. The dial-a-ride problem is concerned with developing a set of routes for a fleet of vehicles carrying customers between specified origins and destinations.

The routing of an AGV on a shop floor encounters conflicts along the arcs and at the nodes (P/D points and path intersections). The analytical studies for dial-a-ride problems (for example, Stein [167] and Jaw et al. [78]) do not account for conflicts,

either along the arcs or at the nodes, hence the distinctive nature of the AGV routing problem. Because of the advantages of bidirectional paths [38,91], and the complex nature of the conflict-free routing problem, almost all the literature on conflict-free routing considers bidirectional guide paths.

Blair et al. [10] consider the optimal routing of AGVs with the objective of minimizing the maximum tour length of an AGV, under the assumption that conflict is negligible. They also assume that the sequence of moves that are assigned to each AGV is a tour. They formulate this problem as a multiple traveling salesman problem. The nodes denote move requests and the arcs define their sequence. The heuristic they propose is composed of two phases: Phase I is a branch and bound approach to find the set of tours that minimizes the sum of all tour lengths; Phase II is a tour improvement process which starts from the solution of Phase I. Fujii and Sandoh [43] find the k th shortest path free of conflicts for each AGV. The value of k is increased until a conflict-free path is found. This path may not be optimal, since the shortest path with conflicts may take less time to travel than a longer path without conflicts. Haines [64] develops a routing algorithm which is applicable to a network where there are no more than two paths leaving any intersection and there exists a path through the network that passes exactly once through each intersection, buffer, and machine.

Krishnamurthy et al. [100] study the conflict-free routing of bidirectional AGVs. They represent the manufacturing facility as a planar graph, formulate the problem as an integer program, and suggest heuristics based on column generation. The master problem of their formulation is a constrained set partitioning problem. The set contains all the possible routes for an AGV. It includes a set of partitioning constraints, and also includes conflict elimination constraints. The variables in the formulation are routes for each AGV. The subproblem determines a lowest cost additional route. Routes generated at the subproblem are added to the appropriate AGV's list of routes. If such a route cannot be found for any AGV, then the procedure is terminated. Computational results show that this model usually provides, in reasonable time, a solution that is within 5% of a lower bound. Langevin et al. [104] consider a manufacturing system served by two AGVs, and describe a dynamic programming approach to find the optimal conflict-free routes. They propose a dominance test to eliminate some states. They also indicate how their approach can be used heuristically in a system with many AGVs.

Chung and Fischer [22] provide a heuristic based on neural networks and simulated annealing to find conflict-free shortest paths in a network. They find that their solution is very close to a global minimum solution for a few example problem instances. Akturk and Yilmaz [3] consider conflict-free AGV routing to minimize the total deviation between delivery or pick up time and the due date. This objective aims for just-in-time delivery and pick up. They present a myopic heuristic which performs better than earliest release time, shortest travel time, and earliest due date dispatching rules.

Case 2: Dynamic job arrivals. One important but very challenging problem is how to accomplish conflict-free routing in real time when jobs arrive randomly, i.e. in a dynamic environment. Kim and Tanchoco [90,91], Dowsland and Greaves [29] and Huang et al. [73] consider a simpler version of this dynamic problem. In their manufacturing system, the AGV routing plan is already established for an initial static job set. They consider the question of how to route a newly arriving job to its desired destination, without conflicts, in the shortest possible time. Kim and Tanchoco [90] address this problem using concepts from graph theory. The free *time windows* of each node are obtained from the AGV routing plan for the initial job set. The intersections of AGV paths and the P/D points define nodes. A new AGV may be routed through a free time window at a node. A time window graph is drawn, with nodes representing free time windows and arcs representing reachability between free time windows. Reachability is defined by: (a) whether two nodes are connected by a physical path, (b) the chronological sequence of the time windows, (c) head on conflict, and (d) catching up conflict. The AGV is routed through the shortest path of the time window graph, which is found in polynomial time by the algorithm of Dijkstra [28].

Kim and Tanchoco [91] provide a myopic strategy in which one vehicle is considered at a time. A vehicle is assigned to the latest move request only when it becomes idle, and a conflict-free route is found by the method of Kim and Tanchoco [90] discussed above. Taghaboni-Dutta and Tanchoco [172] propose another myopic strategy and compare it with the approach of Kim and Tanchoco [90] via simulation. Their myopic strategy is to find the current best edge for the AGV to take, and this incremental path planning is continued to the destination of that AGV. Their results suggest that the approach of Kim and Tanchoco [90] results in better throughput than a myopic strategy. Huang et al. [73] present an approach similar to that of Kim and Tanchoco [90], but they consider the free time windows of the arcs as well as of the nodes in drawing the graph. This eliminates the need to consider head on conflict and catching up conflict separately. Dowsland and Greaves [29] obtain k elementary shortest paths in the *physical network* using the algorithm of Yen [183] with suitable values of k , and introduce delays for AGVs and use spurs wherever conflicts might occur. They calculate the total time, including delays and waiting time along spurs, for each of k shortest paths, and select one with the lowest total time. Since not all possible paths are explored, their approach is only a heuristic. Conflict detection can also be addressed using Petri nets, as in Zeng et al. [185] and Lee and Lin [106]. A standard Petri net is a four-tuple, but in order to accommodate the dynamic aspects of an AGV system, Zeng et al. [185] introduce a colored-timed Petri net which is a seven-tuple. In this Petri net, the vehicle routes are mapped as shared resources. Any conflict is identified by detecting deadlocking in the net, which is known as the Petri net safety problem. Lee and Lin [106] define a special class of Petri nets called attributed Petri nets, in which they model the vehicle flows and present an algorithm to predict and avoid deadlocks. Dhouib and Kadi [27] present an expert system for conflict-free

routing. In their expert system, the conflict management strategy is based on heuristic concepts, including waiting for the AGV in conflict to pass and route modification.

A novel way to avoid AGV conflict is to design the AGV paths so that no conflicts will occur. Bozer and Srinivasan [14, 15] propose a *tandem AGV configuration* which consists of non-overlapping single-vehicle closed loops, with buffers between loops for load transfer. Since no two AGVs will meet in this configuration, AGV conflict is totally avoided. But there are also disadvantages. A load may have to be transported by several AGVs, and hence it may spend a lot of time waiting at the buffers between the loops. The important question is whether the time gained by eliminating AGV conflict is more than the time lost by the jobs waiting at the interloop buffers. Another problem that arises in tandem AGV configurations is which loops (or AGVs) to select when it is necessary to transfer a load from one loop to another. This problem is known as the load routing problem and studied by Lin et al. [113], who propose a two-phase approach. Phase I, based on linear programming, minimizes the total loaded travel time while obtaining the optimal delivery rate of each feasible route. In Phase II, a simulation study validates the results of Phase I. In addressing the load routing problem, Lin and Dgen [114] provide a polynomial time algorithm to find the shortest travel time route, considering the free time windows of AGVs, and using the shortest path algorithm of Dijkstra [28]. The relative performance of tandem and conventional AGV configurations is compared by Bozer and Srinivasan [16] and Choi et al. [21] using simulation, and by Wang and Hafeez [180] using Petri nets, but the results are not yet conclusive. This remains an active research area. Ganesharajah et al. [44] study loop layouts in a deterministic production environment. They identify conditions under which decomposition of a loop layout with multiple AGVs into several nonoverlapping tandem loops each served by a single AGV is worthwhile. The decomposition issue is investigated further by Ganesharajah et al. [45], using an integer programming model and a genetic algorithm.

3.3.2. Joint scheduling

In a complex manufacturing system with a static job set, we may wish to find optimal job and AGV schedules so that global optimality with respect to some objective can be achieved. The joint scheduling problem with the objective of minimizing makespan is actually a nonlinear mixed integer program, as formulated by Bilge and Ulusoy [9]. Let J denote the set of jobs available for processing, and n_j the number of operations of job j , where $n = \sum_{j \in J} n_j$. Let $I = \{1, \dots, n\}$ denote the index set of operations, and $N_j = \sum_{i=1}^{n_j} n_i$. Also, $I_j = \{N_j + 1, \dots, N_j + n_j\}$ is the set of indices in I associated with job $j \in J$, $\bar{I}_i = I \setminus \{h : h \geq i, i, h \in I_j\}$ and $\underline{I}_h = I \setminus \{i : i \leq h, i, h \in I_j\}$. Thus, $\bar{I}_i$ is the index set of operations excluding operation i and succeeding operations of the same job, and $\underline{I}_h$ is the index set of operations excluding operation h and preceding operations of the same job. Also, p_i and the variable c_i represent the processing time and completion time of operation i , r_j is the ready time of job j , and t_i and the variable T_i represent the travel time and completion time of loaded trip i ,

respectively. The travel time of an empty trip starting at the machine performing operation h and ending at the machine performing operation i is denoted by τ_{hi} . The number of AGVs in the system is given by K . Besides c_i and T_i , we also use the following variables:

$$y_{rs} = \begin{cases} 1 & \text{if } c_r < c_s, \text{ where } r \text{ and } s \text{ are operations of different jobs,} \\ 0 & \text{otherwise;} \end{cases}$$

$$x_{hi} = \begin{cases} 1 & \text{if an AGV is assigned for the empty trip between trip } h \text{ and trip } i, \\ 0 & \text{otherwise;} \end{cases}$$

$$x_{oi} = \begin{cases} 1 & \text{if an AGV starts from the I/O station to perform trip } i \text{ as its first assignment,} \\ 0 & \text{otherwise;} \end{cases}$$

$$x_{ho} = \begin{cases} 1 & \text{if an AGV returns to the I/O station after completing trip } h \text{ as its last assignment;} \\ 0 & \text{otherwise.} \end{cases}$$

Given these definitions, the problem can be formulated as follows:

Minimize Z

$$\text{subject to } Z \geq c_{N_j + n_j}, \quad j \in J, \quad (1)$$

$$c_i - c_{i-1} \geq p_i + t_i, \quad i, i-1 \in I_j, j \in J, \quad (2)$$

$$c_{N_j+1} \geq p_{N_j+1} + t_{N_j+1} + r_j, \quad j \in J, \quad (3)$$

$$(1 + H\tau_{rs})c_r \geq c_s + p_r - Hy_{rs}, \text{ and}$$

$$(1 + H\tau_{rs})c_s \geq c_r + p_s - H(1 - y_{rs}), \quad r \in I_j, s \in I_k, \quad (4)$$

where $j, k \in J, j < k$,

$$x_{oi} + \sum_{h \in \bar{I}_i} x_{hi} = 1, \quad i \in I, \quad (5)$$

$$x_{ho} + \sum_{i \in \underline{I}_h} x_{hi} = 1, \quad h \in I, \quad (6)$$

$$\sum_{i \in I} x_{oi} \leq K, \quad (7)$$

$$\sum_{i \in I} x_{oi} - \sum_{h \in I} x_{ho} = 0, \quad (8)$$

$$T_i \leq c_i - p_i, \quad i \in I, \quad (9)$$

$$T_i - t_i \geq c_{i-1}, \quad i, i-1 \in I_j, j \in J, \quad (10)$$

$$T_i - t_i \geq x_{oi}\tau_{o,i-1} + \sum_{h \in \bar{I}_i} x_{hi}(T_h + \tau_{h,i-1}), \quad i, i-1 \in I_j, j \in J, \quad (11)$$

$$T_{N_j+1} - t_{N_j+1} \geq \sum_{h \in \bar{I}_{N_j+1}} x_{h,N_j+1}(T_h + \tau_{ho}), \quad j \in J, \quad (12)$$

$$c_i \geq 0, \quad i \in I, \quad (13)$$

$$x_{hi} \in \{0, 1\}, \quad h \in \bar{I}_i, i \in I,$$

$$x_{oi}, x_{ho} \in \{0, 1\}, \quad i, h \in I,$$

$$y_{rs} \in \{0, 1\}, \quad r \in I_j, s \in I_k,$$

where $j, k \in J, j < k$.

The objective is to minimize the makespan, which is achieved by minimizing the maximum of the completion times of the last operations for all jobs. Constraints (2) and (3) are the precedence constraints for the operations. Constraints (4), where H is a large positive number, are in either-or form and ensure that no two operations that are assigned to the same machine can be processed simultaneously. Constraints (5) and (6) ensure that all the operations are started and finished once. Constraint (7) requires each AGV to enter the system at most once. Constraint (8) keeps the number of AGVs in the system consistent. Constraint (9) states that operation i can start only after the trip to load it has been completed. The next three constraints, (10), (11) and (12), are related to the starting times of the trips. Together, they state that trip i cannot start before the maximum of the completion time of the previous operation and of the empty trip to the previous operation.

This formulation is intractable as we now discuss. Indeed, constraints (1), (2), (3) and (4) together with the integrality constraints and the objective function define a precedence constrained makespan minimization problem, which is known to be unary NP-hard [109]. On the other hand, constraints (5), (6), (7) and (8) define a generic vehicle scheduling problem which is also unary NP-hard [137]. These two subproblems interact through constraints (9) and (10), which when taken together define a time window around T_i : $c_{i-1} + t_i \leq T_i \leq c_i - p_i$.

Due to the intractability of the joint scheduling problem as explained above, the literature contains mainly heuristic approaches. Several researchers address this problem in a hierarchical manner (e.g., first AGV dispatching is addressed, followed by job scheduling), making use of the linkage between these two problems. Bilge and Ulusoy [9] decompose the problem into two subproblems and solve it using an iterative heuristic. The heuristic is based on three components: (a) a procedure that generates machine schedules, (b) a procedure that, given a machine schedule, finds a solution to the vehicle scheduling problem, and (c) an iterative structure that links the two and facilitates the search for a good solution. Procedure (a) is based on three dispatching rules (most work remaining, least work remaining, and shortest processing time).

Procedure (b) is a sliding time window heuristic which checks whether it is possible to accomplish all material handling trips with the given number of vehicles within the time windows resulting from procedure (a). If this is not the case, instead of stopping without a feasible solution, the original makespan is increased, new time windows are constructed, and the heuristic is reapplied. A numerical example using different layouts of four machines highlights the strong relationship between the type of layout and the heuristic performance.

Raman et al. [143] consider the joint scheduling problem with the objective of minimizing the total tardiness. However, they assume that AGVs always return to the I/O station upon completion of their task. This problem is formulated as a zero–one integer program which is similar to a resource-constrained project scheduling problem. They specify a branch and bound solution methodology for this problem, and recommend that this approach be applied using a rolling horizon for the dynamic case where jobs keep arriving in the system.

Jaikumar and Solomon [77] consider job and AGV scheduling in a two-level hierarchical manner for a Japanese manufacturing system, where the jobs are returned to a central warehouse after each machining operation. Their primary objective is maximizing machine utilization and their secondary objective is minimizing total AGV travel time. They also consider minimizing AGV fleet size as a tertiary objective. They fix the AGV arrival times at a machine j to be an ordered set $\{a_{j,1} \leq a_{j,2} \leq \dots\}$ and create a time-space network as follows. The nodes are $(j, a_{j,i})$, $i = 1, 2, \dots$, $j = 1, 2, \dots$. An arc between nodes $(j_1, a_{1,1})$ and $(j_2, a_{2,2})$ in the network exists if it is feasible for the same AGV to travel from machine j_1 to the central warehouse and then to machine j_2 within the time window $[a_{1,1}, a_{2,2}]$. The optimal AGV assignments are found from a minimal chain decomposition of the above network. Each chain corresponds to a sequence of AGV routes covered by the same AGV. The minimal chain decomposition, which achieves minimum fleet size, can be found in polynomial time by solving a related maximal flow problem [152]. They also provide a polynomial time machine sequencing algorithm, which is based on the intuition that jobs should be sequenced on each machine in such an order that an AGV visit to that machine occurs on or before the completion time of the corresponding job.

3.3.3. Dispatching rules

In a complex manufacturing system with many machines, many AGVs and many available paths, the control of the system is a complex task. In particular, when many machines are competing for one AGV, it is necessary to decide which machine should be served. Similarly, when one machine is competing for many AGVs, an AGV must be chosen. Egbelu and Tanchoco [37] call the former problem vehicle initiated and the latter problem work center initiated. The joint problem is known as the AGV dispatching problem. Table 3 provides a comprehensive list of all the vehicle initiated and work center initiated dispatching policies considered in the literature in alphabetical order. The acronyms given in this table are widely used by researchers and

Table 3
Summary of AGV dispatching policies.

Vehicle initiated rules	Acronym	Work center initiated rules	Acronym
First-Come-First-Served	FCFS	Farthest Vehicle	FV
First-Encountered-First-Served	FEFS	First Available Vehicle	FAFS
Largest Queue Size	LQS	Least Cumulative Idle Time	LIT
Longest Inter-arrival Time	LIT	Least Utilized Vehicle	LUV
Longest Travel Time	LTT	Longest Idle Vehicle	LIV
Longest Waiting Time	LWT	Most Cumulative Idle Time	MIT
Maximum Demand	MD	Nearest (Idle) Vehicle	NV or NIV
Maximum Outgoing Queue Size	MOQS	Random Vehicle	RV
Minimum Remaining Outgoing Queue Size	MROQS		
Minimum Work-in-queue	MWQ		
Modified First-Come-First-Served	MFCFS		
Random Work Center	RW		
Shortest Travel Time/Distance	STT/D		
Unit Load Shop Arrival Time	ULSAT		
Vehicle Looks For Work	VLFW		

also in this paper. The literature contains many evaluations of different AGV dispatching policies through a variety of methodologies, as summarized in table 4. Some of the most significant issues and results are now discussed.

AGV dispatching policies can also be classified in other ways. Yim and Linn [184] classify them as push rules and pull rules. A push-based policy selects a job (source) first, and then determines the workstation (destination) to which it should be moved. Once the source and destination are determined, an AGV is chosen to perform the selected load movement, whereas a pull-based AGV dispatching policy first uses a specified rule to select a workstation that can receive jobs. Then, a list of jobs that can be moved to this workstation is identified from the output buffers of other workstations. Finally, a job selection rule is applied to this list to select a job. Klein and Kim [96] distinguish between single-attribute dispatching policies and multi-attribute dispatching policies. Dispatching policies such as LQS and STT consider only one attribute (queue size and travel time, respectively) and hence they are single attribute. A dispatching policy which would consider both of these attributes with predetermined weighting factors (e.g., $0.2 * \text{queue size} + 0.9 * \text{travel time}$) is a multi-attribute policy. Their study shows that multi-attribute policies perform better than single attribute policies, as indicated in table 4.

The performance measures considered in the literature include the maximization of unit load throughput [37]. Yim and Linn [184] consider average output rate as the

Table 4

Summary of AGV dispatching studies.

Author(s)	Policies	Methodology	Main results
Egbelu and Tanchoco [37]	Nearest vehicle Farthest vehicle Longest idle vehicle Maximum outgoing queue size Shortest travel time Longest travel time Minimum remaining outgoing queue Modified FCFS	Simulation	Shop locking phenomenon is identified. All work center initiated rules perform similarly.
Russell and Tanchoco [145]	Largest queue Longest waited load Nearest load Random assignment	Q-GERT simulation	Policy does not influence mean flowtime, machine utilization, AGV utilization, or maximum input queue length. It influences length of output queues.
Cheng [20]	1. First available AGV 2. Most idle AGV 3. Least idle AGV 4. Nearest AGV 5. Farthest AGV	Simulation	Rules 1 and 4 are superior. Rules 2 and 3 are outperformed. Rule 5 performs moderately. AGV speed critically affects performance.
Egbelu [30]	Pull and push rules	Simulation	Pull rules compete with push rules.
Hodgson et al. [71]	Serve station with highest weighting factor	Semi-Markov decision process	Near-optimal rule. Performs better than “Vehicle Looks for Work” rule of Newton [134].
Occeña and Yokota [136]	Maximum demand rule	Simulation	Better than previous rules for JIT systems.
Sabuncuoglu and Hommertzhaim [146]	Four-level hierarchy	Simulation	Better than previous rules.
Sabuncuoglu and Hommertzhaim (1993)	FCFS LQS	Simulation	LQS is better than FCFS
Raju and Chetty [142]	LUV, NV, LIV, RV	Petri net	NV is best, LIV and LUV balance load of AGVs.
Yim and Linn [184]	Push rules Pull rules	Petri net and simulation	Reduce pallet size to avoid shop locking. Push and pull perform similarly.
Mahadevan and Narendran [122]	Least utilized vehicle Nearest idle vehicle First available vehicle Fixed path vehicle dispatch Modified fixed path vehicle dispatch	Simulation	Nearest idle vehicle rule performs best. Fixed path vehicle rule performs badly.
Shang [151]	FCFS STT LQS	Simulation Taguchi methods Response surface method	LQS is better than STT, which in turn is better than FCFS.
Klein and Kim [96]	Single attribute and multiattribute rules	Fuzzy set theory and simulation	Multiattribute rules perform better than single attribute rules.
Bozer and Yen [17]	Modified STT Bidding based dynamic dispatching	Simulation	Both rules perform considerably better than STT.

performance measure. Occeña and Yokota [136] consider transport efficiency and logistic efficiency as the performance measures. They define transport efficiency as the ability to meet target production levels, and logistic efficiency as the ability to enforce low inventory levels at the P/D stations, which is a requirement for the JIT systems they consider. Other performance measures considered in the literature include AGV utilization, machine utilization, mean job flow time, maximum length of queues at P/D points, and mean job tardiness.

A major problem that can arise from not using a proper dispatching policy is the phenomenon called *shop locking*. Shop locking occurs when dispatching policies are functions of distance measures (e.g., shortest travel time, longest travel time). In this case, it may happen that some departments are poorly served by these policies. Note that the performance of distance based dispatching policies is strongly linked to the facility layout of the system, i.e., the location of machines and P/D points. Eventually, the output queues may become full and subsequently machines are blocked. This is followed by the blocking of vehicles making deliveries in those departments, as the input queue grows to its maximum capacity. This chain reaction may propagate to all vehicles in the shop, causing a complete seizure of material flow. The phenomenon of shop locking is well explained by Egbelu and Tanchoco [37]. The way to avoid shop locking is to decrease the number of jobs in the system, as explained by Yim and Linn [184]. Having a large number of jobs in the system increases the possibility of shop locking, because of excessive congestion and traffic on the shop floor. However, having too few jobs in the system tends to decrease the average output rate. Another method of congestion control, as proposed by Taghaboni-Dutta and Tanchoco [171], is to limit the number of unit loads in the system to a prespecified value and to select a new unit load to enter the system based on the congestion of the machine to which this unit load is headed first.

4. Future research issues

We begin with a discussion of AGV flowpath design issues. Most heuristics, for example CRAFT, that are used to lay out departments of a manufacturing facility consider only the distances between centroids of departments for estimating total material flow. This distance does not reflect the actual material movement between two departments in the case of AGVs, due to the many different flowpaths that AGVs may take during their operation. Thus, there is a strong need to integrate the facility layout and the flowpath design. Montreuil [133] formulates this integrated problem as a mixed integer program, but does not suggest a solution approach. Banerjee and Zhou [7] integrate departmental layout design with the design of a single loop AGV path passing through all the departments. They suggest the use of genetic algorithms to solve the integrated problem. Afentakis [1] provides a heuristic procedure for arranging machines in a loop layout considering exact material movement distances, with the objective of minimizing the number of machines that the jobs cross during their

manufacturing process. Heragu and Kusiak [70] address the machine layout problem for a bidirectional single-path, single-AGV system. They arrange the machines in a line according to the frequency of trips to be made by the AGV. They explicitly consider the AGV velocity, which is a nonlinear function of time, in arranging the machines. Similar studies for ladder type layouts, segmented AGV layouts, and complex network layouts need to be undertaken.

Design issues for a single loop that passes through all departments are studied in the literature. But the design of many single loops that operate in tandem, as proposed by Bozer and Srinivasan [15], is an area needing further exploration. The related problem of integrating the single loops (in tandem) and the facility layout (i.e., the location of the workstations), as discussed in the preceding paragraph, is another interesting open area. As explained by Bozer and Srinivasan [16], the benefits of tandem AGV configurations will be more fully realized when this integration has been achieved. Faraji and Batta [39] provide a simulation based procedure to form single vehicle loops. However, analytical methods are not yet available.

In the area of flowpath design when departmental layout is given, very few studies consider adding shortcuts and spurs, which have the potential to reduce the conflict time significantly. Sharp and Liu [153] and Dowsland and Greaves [29] consider these approaches, which can be applied to a variety of flowpath design issues.

In the area of fleet size design for single loops, analytical models can be developed, as in Ganesharajah et al. [45]. The fleet size design of multiple-load carrying AGVs has been studied by Leung et al. [110], Ozden [138], and Johnson and Brandeau [82]. Since multiple-load carrying AGVs decrease the number of AGVs needed, thereby reducing congestion, this seems to be an appealing alternative. Analytical studies are needed to evaluate such alternatives.

The use of bidirectional AGVs may substantially decrease the length of the flow path needed. When two-way travel is permitted, the total AGV travel time required may decrease, resulting in a reduced fleet size requirement. These advantages of bidirectional AGVs are studied by Egbelu and Tanchoco [38], and Kim and Tanchoco [91], as discussed in section 2.1.4. Even though the control of such systems is expensive, ongoing developments in control systems are likely to make implementation economically feasible in the near future. There appear to be no studies on flowpath design or fleet sizing that consider the use of bidirectional AGVs, which remains an area needing investigation. Another interesting related topic is how to design a mixed flowpath that combines the advantages of unidirectional and bidirectional paths.

In the area of conflict avoidance of bidirectional AGVs, the subproblem of the column generation method given by Krishnamurthy et al. [100], which generates shortest path routes, can be integrated with the Petri net based conflict-detection model of Zeng et al. [185], thereby eliminating the need to use the time-consuming master problem to find conflict-free routes. This integrated model may find conflict-free routes faster, thus making real-time applications feasible. Shiizuka and Suzuki [154] present a Petri net model for AGV networks. This model may be extended to incorporate

conflict detection. The effectiveness of tandem AGV configuration for large flowshop loops is investigated by Ganesharajah et al. [45]. They identify cases where the use of a tandem AGV configuration always increases throughput rate. This result motivates several further investigations.

The existing studies on conflict-free routing do not consider the issue of how to update the current routing plan when jobs arrive dynamically. That is, those studies assume that the current plan is fixed and find ways to route the newly arriving jobs. This approach is clearly suboptimal. Finding ways to update the current plan as and when new jobs arrive is an important challenge that can offer increased throughput.

The ladder type of AGV network (figure 3) is implemented, for example, at FIAT for automobile assembly [101]. This type of network is used where a number of alternative processes have to be serviced by AGVs. The AGVs may visit the stations in different orders, depending on the product being produced. Design and operational issues in such AGV layouts should be investigated. Since these layouts are not very complex, we believe that analytical study of some issues should be possible.

Cyclic scheduling, as practiced in the Toyota production system, has many advantages to offer, as discussed by Hall [68]. Here, parts are produced repetitively and in proportion to the demand, thus reducing the inventory of finished parts [132]. In the area of cyclic scheduling, analysis is needed even for single line and single loop AGV systems. Some preliminary results in this area have been obtained by Ganesharajah [44] and Ganesharajah et al. [45].

The practical benefits of AGV scheduling will be increased by identifying why an AGV dispatching policy performs well under certain conditions and poorly under others. This information can be incorporated in the layout design (for example, in the location of the machines and P/D points) of the manufacturing system. Further information that AGV system designers need is how different objective measures vary with AGV fleet size under various dispatching policies. With this information, managers should be able to select the fleet size and appropriate dispatching policies. Only a few authors, for example Jaikumar and Solomon [77], have so far addressed this integrated scheduling and design problem.

One problem which has received only recent attention is the determination of home location of AGVs after each delivery. As discussed above, Egbelu [33] and Chang and Egbelu [19] consider this problem in a loop layout. Majety and Wang [123] integrate this problem with the flowpath design problem in a complex network. Similar studies for line layouts and segmented loop layouts are needed. The related question of whether locating the AGV at the point of last delivery, or moving it to predefined terminals after each delivery, is advantageous, has not been answered even in simpler layouts. This is an important issue that can offer improved operation of AGV systems. Analytical studies should be possible in simpler layouts, and simulation studies are needed for complex layouts.

The important issue of joint optimization of job and AGV schedules has so far been discussed by only a few authors [9, 77, 143]. Since this problem is a nonlinear

mixed integer program, it would be useful to tailor the latest nonlinear programming techniques to find exact optimal solutions. Because there are many locally optimal solutions in such problems, the usefulness of heuristic approaches such as simulated annealing [177], genetic algorithms [54] and tabu search [51] should also be explored.

The design and operational issues that arise in free-ranging AGVs need to be addressed as the technology develops. The limitations of the virtual flow path design model as discussed by Gaskins et al. [49] should be investigated. Taghaboni and Tanchoco [170] develop a LISP based controller for free-ranging AGV systems. This controller performs dispatching, scheduling, and route selection decisions for a fleet of free-ranging AGVs. However, since such expert systems approaches often identify feasible but not optimal solutions, several challenging problems remain for operations researchers.

Throughout this paper, we mention the need to integrate design and operational issues. We now need to reemphasize this point. The facility layout problem requires knowledge of the flowpath to estimate travel distances between departments. Effective flowpath design requires information about the travel time between departments, which should include time lost due to AGV conflicts. However, AGV conflicts can be measured only if the dispatching policies and fleet size are known. The fleet size design problem requires knowledge of AGV dispatching policies to estimate empty vehicle travel and conflicts. Thus, it is inefficient to consider these problems independently. To address this concern, researchers including Maxwell and Muckstadt [128] and Johnson and Brandeau [82] suggest an iterative procedure in which one design is assumed and operational problems are studied. The operational results are used as input for a new design, and this procedure is repeated. Sharp and Liu [153] develop a procedure of this type. They consider the delay due to conflicts and empty vehicle travel to determine fleet size analytically and also to identify the optimal way to add shortcuts and spurs in a given AGV flowpath. A SLAM simulation model validates the analytical results. In a similar iterative procedure, Johnson [79] demonstrates that including empty travel in a flowpath design model results in a solution with substantially reduced total empty travel, which helps to reduce the fleet size requirement. However, this iterative procedure is time consuming. Further analytical studies are needed to develop insights about how these issues are related, and to provide rules for better design and operation of AGV systems. The evolving philosophy of *concurrent engineering* [102] calls for making such linkages between issues. In keeping with the concurrent engineering philosophy, Kusiak and He [103] provide rules for the design of products so that scheduling becomes easier in a general manufacturing context. Similar studies are needed for designing AGV systems so that operational issues, including scheduling, become easier to address.

5. Conclusions

The widespread and increasing use of AGVs for material handling in industry has generated a substantial body of literature on related design and operational issues.

This paper summarizes the optimal and heuristic procedures available for flowpath design, determining the AGV fleet size, and for operational issues such as job scheduling, AGV dispatching and conflict-free routing. Most real-world applications of AGV systems define problems that are intractable. Neighborhood search techniques like tabu search or genetic algorithms can be applied to the flowpath design problem. Existing models roughly estimate the empty vehicle time and AGV conflicts in determining the fleet size. More precise models should be developed by identifying parallels in the transportation literature and in network flow models. In the scheduling studies described, the usual objective is minimizing the makespan. Other objectives, for example minimizing the lateness relative to target times, minimizing the flowtime, and minimizing the AGV utilization, should also be explored. Since JIT is a popular management philosophy in modern AGV-served manufacturing systems, the two objective problem of minimizing lateness and minimizing earliness arising in JIT systems is also useful to study. The methods used in the literature to study most of the problems are analytical, heuristic or simulation-based. Modern tools, including Petri nets, neural network modeling, and artificial intelligence, have considerable potential to improve the state of knowledge in this area. Since all the above issues are inter-related as we discuss, integrated decision making for better design and better operation remains an important challenge.

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